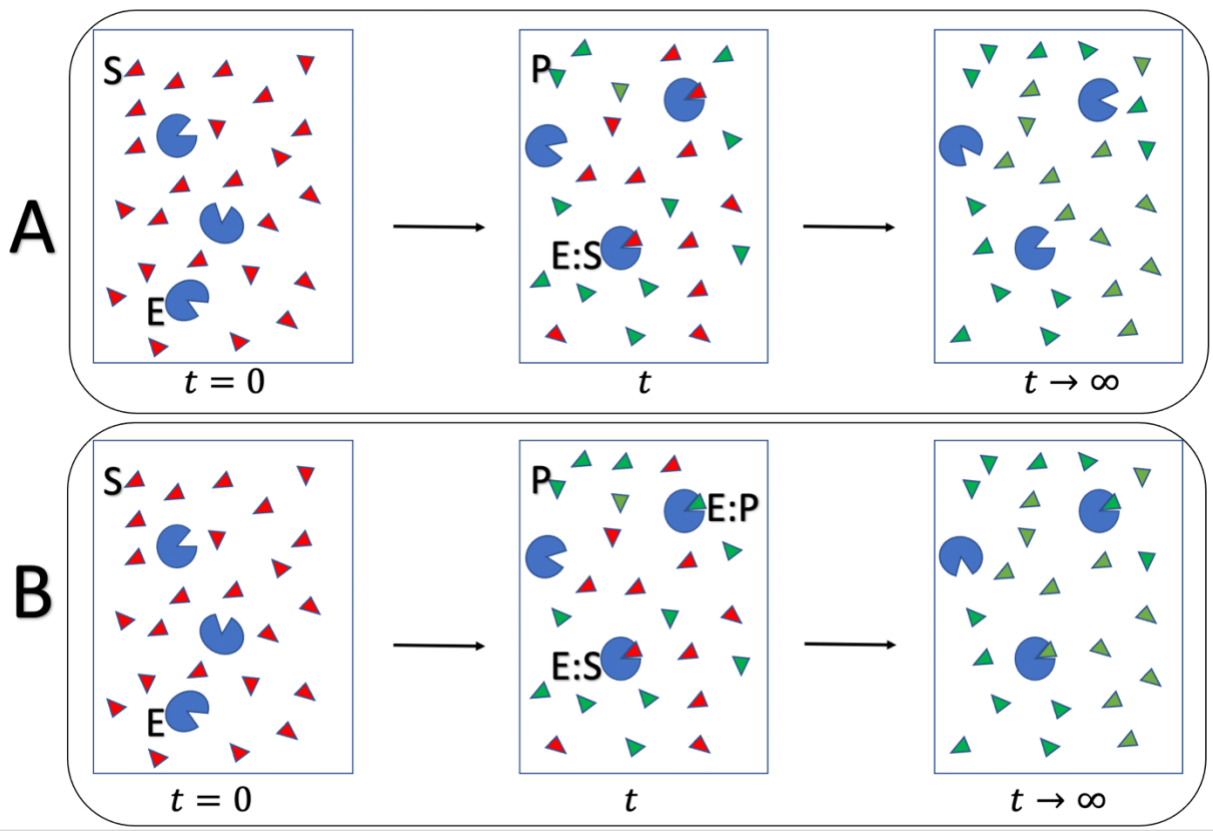
Physical Chemistry CUME
April 9, 2021

Schematics below show the snapshots of two different catalytic enzyme reactions. The initial concentration of substrate (S) is $[S]_0$ and the initial concentration of enzyme (E) is $[E]_0$. Both $[S]_0$ and $[E]_0$ are known. Both reactions have an intermediate, that is the E:S complex. In reaction B, however, the E:P complex is also stable.

1. Based on the schematics below, write down the reaction mechanism associated with reaction A (top). [10 points]
2. For reaction A and within the steady state approximation, express the rate of product formation ($\nu = \frac{d}{dt}[P]$) in terms of the concentration of substrate ($[S]$). In your expression, you may use $[E]_0$ as well as the rate constants of the elementary steps involved in the reaction mechanism. [40 points].
3. Repeat Problems 1 for Reaction B (bottom). [10 points]
4. For reaction B and within the steady state approximation, express the rate of product formation ($\nu = \frac{d}{dt}[P]$) in terms of the concentrations of substrate ($[S]$) and product ($[P]$). In your expression, you may use $[E]_0$ and $[S]_0$ as well as the rate constants of the elementary steps involved in the reaction mechanism. [40 points].